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Expt. No.

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Page No.

23

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Estimation of Iron Content of the given Solution Using Potentiometer.

Aim:-

To estimate the amount of ferrous ion present in the given solution potentiometrically. A standard solution of potassium dichromate of strength 0.2N is provided.

Principle:-

Potentiometric titrations depend on measurement of EMF between reference electrode and an indicator electrode. When a solution of ferrous ion is titrated with acidified potassium dichromate solution, the following redox reaction takes place.

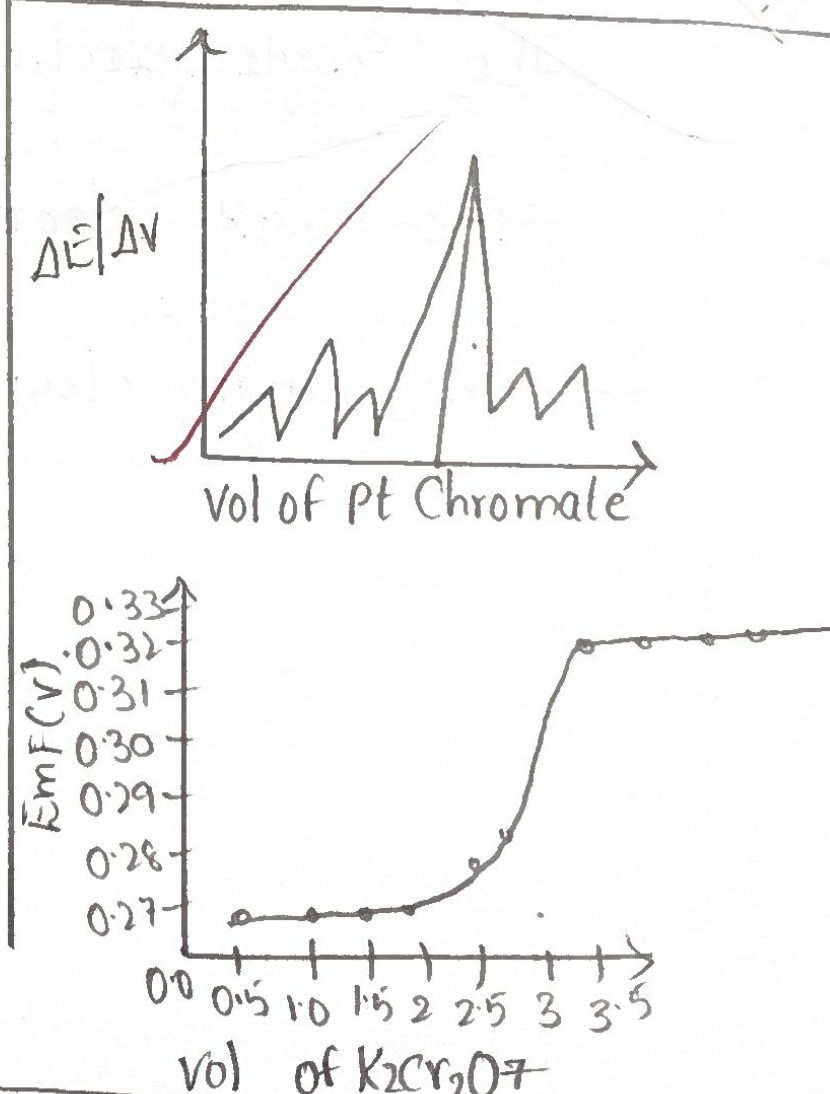


During this titration, Fe^{2+} is oxidised to Fe^{3+} and its concentration increases. At the end point, there will be a sharp rise in the EMF value. This is called "Sweeping potential". It indicates complete formation of Fe^{3+} ions at the end of redox reaction.

Pt, Fe^{2+} , Fe^{3+} // KCl, HgCl_2 (s) Hg.

A graph is drawn between EMF measured against the volume of potassium dichromate added. The point is noted from graph.

Short procedure:

S. NO	Contents.	titration.
1	Burette solution.	std $K_2Cr_2O_7$
2	Pipette solution.	ferrous ion solution.
3	Additional solution.	20ml distilled water 10ml dil. H_2SO_4 .
4	Electrode / cell used.	pt electrode / Calomel electrode.
5	Instrument used.	Potentiometer
6	End point.	EMF increases and sudden increases.
7	Eq. wt of Fe^{2+}	55.85
8	Model graph.	

Procedure:

Transfer the given ferrous solution into 100ml Standard flask and make it upto the mark using distilled water upto to mark.

Pipette out 20ml of ferrous solution in a 100ml Clean beaker. Add 10ml of dil H_2SO_4 and 10ml of distilled water to it. Dip in platinum electrode and calomel electrode. In to solution and connect to a potentiometer. Then add potassium dichromate. Intermis of 1ml from the burette to the solution in the beaker and stir it well. For each addition of the solution from the burette, note the emf of the cell using potentiometer continue the titration atleast six reading after the "Sweeping Potential".

TABLE 1

Volume of $K_2Cr_2O_7$ (ml)	EMF (volts)
1	0.418
2	0.431
3	0.451
4	0.499
5	0.678
6	0.710
7	0.718
8	0.724
9	0.728
10	0.739
11	0.731
12	0.733
13	0.715
14	0.728
15	0.730

Plot a first derivative graph by taking $\Delta E / \Delta V$ versus volume $K_2Cr_2O_7$. The strength of ferrous solution and the amount of ferrous ion present are calculated from the end point



Calculation:-

$$\text{Volume of } K_2Cr_2O_7 = \underline{12} \text{ ml (from graph)}$$

$$\text{Strength of } K_2Cr_2O_7 = \underline{0.2 N} \quad \emptyset$$

$$\text{volume of ferrous ion } (V_2) = 20 \text{ ml}$$

$$\text{strength of ferrous ion } (N_2) = (V_1 \times N_1) / V_2$$

$$N_2 = (V_1 \times N_1) / V_2$$

$$\frac{4 \times 0.2}{20} = \underline{0.04 N}$$

The strength of the ferrous is found is found to be $\underline{0.04 N}$.

Amount of the ferrous ion present in 1 litre of the given Solution

$$= \text{strength of strong acid} \times \text{eq. wt}$$

$$= \underline{0.04} \times 55.85$$

$$= \underline{2.234} \text{ g/l}$$

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Result:-

The amount of ferrous ion present in the
given solution is = 2.234 g/l.

TABLE 2

Volume of K ₂ CrO ₇	EMF	ΔE	ΔV	$\Delta E/\Delta V$
0	0.407	0.011	1	0.011
1	0.418	0.013	1	0.013
2	0.431	0.020	1	0.020
3	0.451	0.048	1	0.048
4	0.499	0.179	1	0.179
5	0.678	0.021	1	0.021
6	0.699	0.011	1	0.011
7	0.710	0.008	1	0.008
8	0.718	0.006	1	0.006
9	0.724	0.004	1	0.004
10	0.728	0.002	1	0.002
11	0.730	0.001	1	0.001
12	0.731	0.002	1	0.002
13	0.735	0.002	1	0.002
14	0.735	0.002	1	0.002
15	0.728	0.003	1	0.003